

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi

End-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
12th December 2024

Time: 3 hours

Total Marks: 50

Answer all questions. ($5 \times 10 = 50$ marks)

Q 1. Use Newton's forward difference formula for the given values to find $f(0.5)$ and $f(1.5)$

x	0	1	2	3	4
f(x)	1	7	23	55	109

Q 2. Find the solution of the system of equations

$$2x + y + 2z = 10$$

$$x + 2y + z = 8$$

$$3x + y - z = 2$$

using the Gauss-Jordan method.

Q 3. Find the interval in which the positive roots of the equation $x^2 + 5 = e^x$ lie. Find the smallest positive roots of the equation using Newton-Raphson method.

Q 4. Use the Runge-Kutta method to find approximate values of the solution of the given initial value problem at the points $x_i = x_0 + ih$, where x_0 is the point where the initial condition is imposed and $i = 1, 2$

$$y' = 2x^2 + 3y^2 - 2$$

$$y(2) = 1;$$

Assume the step length $h = 0.05$.

Q 5. Compute

$$I = \int_1^2 \frac{1}{x} dx$$

using trapezoidal and Simpson's rules with $n = 10$, $a = 1$, $b = 2$. Compare with exact solution.